WORKED SOLUTION

STEP I 1995, Question 8

Pure • Linear Differential Equations

STEP I 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q8

The key idea

The unknowns are three functions, but at each fixed x they are just three numbers $f(x)$, $g(x)$, $h(x)$. Substituting the three given solutions into $(*)$ gives three linear equations in these three numbers, which we solve as simultaneous equations. For the second paragraph, the left-hand side of $(*)$ is *linear* in y, so the result for a combination of x, 1 and x^{-1} follows from what each piece produces separately.

Solution

Finding f, g and h

Write Ly for the left-hand side $y'' + f(x)y' + g(x)y$, so $(*)$ says $Ly = h(x)$. We substitute each solution in turn.

- $y = 1$: here $y' = y'' = 0$, so $g(x) = h(x)$.
- $y = x$: here $y' = 1$, $y'' = 0$, so $f(x) + x\,g(x) = h(x)$.
- $y = x^{-1}$: here $y' = -x^{-2}$, $y'' = 2x^{-3}$, so

$$\frac{2}{x^3} - \frac{f(x)}{x^2} + \frac{g(x)}{x} = h(x).$$

Using $h = g$ in the second equation gives $f(x) = (1-x)\,g(x)$. Putting both of these into the third equation and multiplying through by x^3 (allowed, as $x > 0$):

$$\begin{aligned} 2 - x(1-x)\,g(x) + x^2\,g(x) &= x^3\,g(x) \\ 2 = g(x)\left(x^3 - 2x^2 + x\right) &= g(x)\,x(1-x)^2. \end{aligned}$$

Since $0 < x < 1$, the factor $x(1-x)^2$ is non-zero, so we can divide by it. Therefore

$$g(x) = h(x) = \frac{2}{x(1-x)^2}, \qquad f(x) = (1-x)\,g(x) = \frac{2}{x(1-x)}.$$

At each x we solved the three equations step by step with no free choices, so these functions are the only ones that work. The restriction $0 < x < 1$ keeps us away from $x = 0$ and $x = 1$, where they blow up.

So the equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{2}{x(1-x)}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{x(1-x)^2}y = \frac{2}{x(1-x)^2}.$$

When is $ax + b + c/x$ a solution?

The operator L is linear: $L(\alpha u + \beta v) = \alpha\, Lu + \beta\, Lv$ for constants α, β, because differentiation is linear and f, g multiply y' and y only. From the first paragraph,

$$L(x) = L(1) = L(x^{-1}) = \mathrm{h}(x).$$

Hence, for $y = ax + b + c/x$,

$$Ly = a\,L(x) + b\,L(1) + c\,L(x^{-1}) = (a + b + c)\,\mathrm{h}(x).$$

Ask yourself Why can we conclude $a + b + c = 1$, rather than just "$a + b + c = 1$ at some x"?

We need $(a + b + c)\,\mathrm{h}(x) = \mathrm{h}(x)$ for all $0 < x < 1$, that is, $(a + b + c - 1)\,\mathrm{h}(x) = 0$. But $\mathrm{h}(x) = 2/\big(x(1-x)^2\big) > 0$ on this interval, so we can divide by it. So $y = ax + b + c/x$ is a solution of $(*)$ if and only if

$$a + b + c = 1.$$

Checks

- Taking $(a, b, c) = (1, 0, 0)$, $(0, 1, 0)$ or $(0, 0, 1)$ recovers the three given solutions, and each has $a + b + c = 1$.
- Directly: with $x = \frac{1}{2}$ we get $\mathrm{f} = 8$, $\mathrm{g} = \mathrm{h} = 16$. For $y = 1/x$, $y = 2$, $y' = -4$, $y'' = 16$, and $16 - 32 + 32 = 16$, as required.
- The computer checked all three solutions exactly at 49 rational points in $(0, 1)$, and confirmed $Ly = (a + b + c)\mathrm{h}$ for 2000 random (a, b, c, x).

Common slips

- Writing $y'' = -2x^{-3}$ for $y = x^{-1}$: two minus signs multiply to give $+2x^{-3}$.
- Answering $a + b + c = 0$. That is the condition for $ax + b + c/x$ to solve the *homogeneous* equation $Ly = 0$; here the right-hand side is h, not 0.

Beyond STEP

This is the structure of solutions of a linear differential equation: the difference of any two solutions of $Ly = \mathrm{h}$ solves $Ly = 0$. Here $x - 1$ and $x^{-1} - 1$ solve the homogeneous equation, and for a second-order equation two independent solutions give all of them. So $1 + \alpha(x - 1) + \beta(x^{-1} - 1)$ is the general solution, which is exactly the family with $a + b + c = 1$. The Wronskian makes "independent" precise. Boyce and DiPrima's *Elementary Differential Equations* covers all of this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.